

Geoinformatics Project

Cycle slip simulation

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October 2025

1 Introduction

This project focuses on simulating and analyzing satellite observations from the ground station perspective, with particular emphasis on cycle slip detection. By implementing a complete satellite orbit propagation model, we simulate the position of a satellite over a 24 hours period and analyze its visibility and observability from a specific ground station located in Toamasina, Madagascar.

The foundation of our work lies in the accurate computation of satellite trajectories using Keplerian orbital mechanics. Starting from almanac data containing orbital elements, we propagate the satellite's position through successive transformations—from the Orbital Reference System (ORS) to the International Terrestrial Reference Frame (ITRF)—accounting for time-dependent variations in the argument of perigee, inclination, and right ascension of the ascending node. This approach enables us to generate precise three-dimensional coordinates at one-second intervals throughout the entire day.

A critical component of our analysis involves determining the observability windows during which the satellite is visible from the Toamasina ground station. We compute elevation and azimuth angles for each epoch, identifying periods when the satellite rises above the local horizon. These visibility periods are essential for understanding the temporal distribution of available observations and the geometric constraints that affect positioning accuracy.

2 Signal simulation

2.1 Satellite parameters

To simulate the phase observation we used the almanac of the satellite SVN 63 for the 28th November 2016. These are the values used to compute the satellite orbit:

- $\sqrt{a} = 5.153650835037e^3$ [m], square root of semi-major axis
- $e = 3.841053112410e^{-3}$ [m], eccentricity
- $M_0 = 295004883409$ [rad], mean anomaly
- $\Omega_0 = -2.241692424630e^{-1}$ [rad], longitude of the ascending node
- $\dot{\Omega} = -8.386063598924e^{-9}$ [rad/s], rate of node's right ascension
- $i_0 = 9.634782624741e^{-1}$ [rad], inclination
- $\dot{i}$ [rad/s], rate of inclination angle
- $\omega_0 = 9.419793734505e^{-1}$ [rad], argument of perigee
- $\dot{\omega} = 0$ [rad/s], rate of argument of perigee

2.2 Phase simulation

This is the formula we used for the simulation of the phase observation:

$$\Phi = \rho * 1000 + t^s * c + \epsilon / 1000 + N * \lambda_{wl}$$

where:

- Φ is the phase observation
- ρ is the distance from the satellite to the user in kilometers
- t^s is the satellite clock offset
- c is the speed of light in the vacuum
- ϵ is a random gaussian noise in meters
- N is the integer ambiguity
- λ_{wl} is the wavelenght of widelane (0.5 meters)

2.2.1 Distance

To compute the distance from the satellite to the ground station we computed the position of the satellite in ITRF starting from the almanac values and calculated the euclidean distance in three dimensions.

2.2.2 Clock offset

To simulate the clock offset of the satellite we used these three parameters

- $dt_0 = -7.661711424589e^{-5}$, satellite clock offset
- $dt_1 = -3.183231456205e^{-12}$, satellite clock drift
- $dt_2 = 0$, satellite clock drift rate

2.2.3 Noise

We added a gaussian noise in meters following the distribution: $\mathcal{N}(0, 1)$

2.2.4 Integer ambiguity

Integer ambiguity was set to 100 for all epochs.

2.2.5 Others parameters

We decided to neglect the ground station's clock offset, the ionospheric and tropospheric corrections.